

AI Innovation for Global Economic Leadership: The Role of Science Gateways, AI Infrastructure, and Research Software Engineers

Answer to the Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

By the

CI CoE: SGX3 - A Center of Excellence to Extend Access, Expand the Community, and Exemplify Good Practices for CI Through Science Gateways
NSF Award # 2231406

Authors:

Sandra Gesing, (Principal Investigator)

Maytal Dahan (Co-Principal Investigator)

Claire Stirm (Co-Principal Investigator)

Linda Hayden (Co-Principal Investigator)

Janae Baker (Community Project Manager)

This document is approved for public dissemination, and the document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

1 Executive Summary

This response addresses the Request for Information (RFI) published by the Federal Register on February 6, 2025, on developing a comprehensive Artificial Intelligence (AI) Action Plan. The United States (U.S.) has been a pinnacle for economic growth and continues to drive as a world leader towards AI enhancements. In the U.S., leading institutions in collaboration with industry have been exploring the strategic advancement of AI. The further development of AI in the U.S. is critical to the country's economic expansion, national security, and global interests. Our organization is committed to providing actionable insights to strengthen the nation's leadership in AI research, infrastructure, and workforce development while ensuring secure, efficient, and responsible AI deployment.

A strong AI Action Plan must prioritize access to high-performance computing (HPC), scalable AI model development, and robust data-sharing frameworks. It should also address key infrastructure challenges such as hardware and chip development, data center expansion, energy-efficient AI computing, cybersecurity, AI model assurance, and regulatory modernization. Expanding and modernizing AI infrastructure is essential to supporting these priorities by providing centralized access to computing resources, AI tools, and open-source development frameworks, which will accelerate research and industry applications.

Our response highlights the role of AI infrastructure in democratizing AI research and computing access. The National Artificial Intelligence Research Resource (NAIRR) Infrastructure Supplement, conducted by SGX3 – the NSF Center of Excellence for

Science Gateways [1], gathered community input on AI infrastructure requirements through an online survey with over 1,000 respondents, eight focus groups with more than 60 participants, and a two-day workshop attended by 85 participants. These different formats reveal that researchers prioritize access to shared AI resources for data sharing, computational tools, and collaborative projects. AI researchers rely on personal or institutional computing resources. From our deep understanding of their needs, they shared a strong interest in scaling and connecting their computational resources and workflows to connect with other experts in the U.S. Outcomes from our survey also underscored the demand for AI tools from leading institutions and industry, such as Jupyter and PyTorch, to be in a connected environment for the AI community to elevate access for all and cross-share expertise. Simplifying access to AI infrastructure, improving research portals, and fostering strategic partnerships will be crucial for increasing adoption and impact.

As a mechanism for connecting expertise in AI to these resources, the importance of science gateways and Research Software Engineers (RSEs) is underscored in optimizing AI research workflows. Science gateways are user-friendly interfaces connecting anyone seeking to use complex computational tools in a simplified user experience. Science gateways support researchers, educators, and policymakers in engaging with AI technologies and HPC through these interfaces. Additionally, they support curated access to data and models and tie them together through website-like interfaces. RSEs are vital in designing, maintaining, and securing AI research infrastructure. These expert engineers ensure the community can use these science gateways and models and create website-like interfaces to enable access. RSEs focus on making science gateways usable, reproducible, scalable, and aligned with security priorities. Increased investment in AI-driven science gateways and RSEs will strengthen the U.S. position in AI.

To advance U.S. AI leadership, we recommend expanding AI infrastructure through science gateways, investing in RSEs to support AI software infrastructure, enhancing AI security and computational efficiency, and strengthening collaborations between academia, industry, and government. Additionally, an interconnected AI portal with streamlined connections to industry and research systems, sharing educational resources, and growing community adoption of AI is essential to advance AI solutions.

We are eager to contribute our expertise to shaping a powerful AI Action Plan and supporting the research, development, and deployment of these solutions.

2 Key Recommendations

To ensure U.S. leadership in AI, we recommend strategically expanding a national AI infrastructure as a federated AI resource. The U.S. can innovate through streamlined services by integrating science gateways, RSEs, and AI agents. These elements will provide the necessary infrastructure to support AI-driven research, development, and deployment at scale.

1. **Strengthen the national AI solutions through a centralized AI science gateway.**

An AI science gateway should be developed as an interoperable AI platform, connecting HPC, data resources, and AI tools in one web-accessed system.

Science gateways should be user-friendly access points to streamline computing allocations, data sharing, and AI model training. AI agents should be incorporated to automate workflows, optimize resource allocation, and provide decision-support tools for researchers, policymakers, and industry partners.

2. Invest in RSEs to enhance AI infrastructure.

A well-trained workforce is essential to AI leadership. Thus, federal investment should be made in training RSEs to design, optimize, and maintain AI research infrastructure. Enabling further education of RSEs will support building AI model assurance frameworks, developing open-source AI tools, and ensuring the usability of an AI computing ecosystem.

3. Enhance computing power and secure AI infrastructure.

The national AI infrastructure should continue supporting access to HPC resources. Growth for more powerful resources should also be explored to enable more dedicated GPUs for model training and secure cloud environments. Public-private partnerships should be leveraged to ensure cost-effective access to cloud services. Additionally, Investing in AI capabilities for national security applications such as data privacy, cybersecurity, and AI safety standards should be strengthened to protect against attacks.

4. Expand AI education, training, and industry collaboration.

Gaps in AI adoption and workforce readiness exist. We recommend increasing training and standardization efforts in the research and education communities. An open AI portal would strengthen AI adoption in all industries. This portal should include educational and training resources such as tutorials, workshops, and AI literacy programs tailored to different interests. Industry collaboration should be expanded to accelerate AI deployment in defense, healthcare, finance, and other critical sectors, fostering economic growth while ensuring the responsible integration of AI systems.

5. Optimize the usability of an AI portal with AI-driven support systems.

Given the different needs of the community, expand an AI portal to be used by all audiences - from those adopting AI to those experts seeking to extend AI for novel solutions. An AI portal should balance ease of use for new users with advanced capabilities for AI specialists. To enable a science gateway that meets the needs of multiple audiences, AI agents should be deployed to assist with onboarding, training, resource allocation, automate data management, and provide real-time guidance on using AI tools. The portal should also incorporate feedback-driven design updates, ensuring it evolves based on user needs.

6. Establish AI procurement and regulatory guidelines.

AI policies and regulatory frameworks should be developed for adoption. These policies should incorporate open-source AI development, promote technical and safety standards, and reduce barriers to support novel deployment. Additionally, export controls and international collaboration agreements should be strengthened to safeguard AI intellectual property while fostering cooperative AI research with allied nations.

Through these efforts, the U.S. will be a leader in the foundation of AI innovation, accelerate economic and scientific breakthroughs, and maintain global leadership in AI.

3 Introduction

Artificial intelligence is rapidly advancing across all industries, driving innovation across healthcare, finance, defense, and manufacturing. Breakthroughs in deep learning, generative AI, and large-scale foundation models have enabled powerful applications. For example, AI agents have simplified and streamlined real-time decision-making and complex data analysis. These advances have been fueled by increased computing power and resources. However, access to these resources remains uneven, limiting AI innovation outside well-funded institutions and major technology firms.

A national AI infrastructure should be designed to bridge this gap by expanding access to AI and fostering widespread innovation in the U.S... researchers, educators, and small enterprises struggle to secure the computational resources, datasets, and AI tools needed for cutting-edge AI development. A national AI infrastructure would provide a shared platform. Supporting innovation by tapping into the nation's compute resources, pre-trained AI models, and large-scale data repositories. Thus enabling broader participation in AI research and application.

Science gateways and AI-powered research portals are key to the national AI infrastructure's success. These gateways will lower technical barriers, making AI adoption more accessible across disciplines. RSEs will play a key role in building and developing AI-aligned science gateways, ensuring that the AI models on these science gateways are optimized. The expertise of the RSEs in software development and data engineering will be essential for supporting AI research at scale.

AI agents will further enhance adoption. Automating routine development tasks, optimizing resource allocation, and providing decision-support tools are all proven ways these agents will assist the community of portal users and developers in streamlining key tasks. For example, these tasks would include data processing, model selection, and workflow automation for AI researchers, significantly improving efficiency and accelerating innovation. Elevating these tasks would enable a shorter time for results. Leading to shortened time towards innovation.

By investing in AI infrastructure through science gateways, leveraging RSEs, and extending the use of AI agents, the U.S. can secure its position as a global leader in AI.

4 Current State of AI in the U.S. and Globally

AI has evolved from a conceptual technology to a transformative force reshaping industries, scientific research, and daily life. Driven by continued advancements, AI is now at the forefront of international attention and innovation. We see AI in all sectors. However, challenges must be addressed as AI systems become more powerful to ensure responsible and effective deployment.

4.1 Current National AI Infrastructure

NAIRR is a federal initiative to democratize access to AI resources across academia, industry, and government. Designed to provide equitable access to cutting-edge AI tools, computing power, and high-quality datasets, NAIRR seeks to accelerate AI research and innovation while ensuring that a broad range of users—beyond large

technology firms—can participate in AI advancements. NAIRR envisions a federated, secure, and interoperable AI infrastructure that enables researchers to develop, test, and deploy AI models. The NAIRR Pilot Portal (see below) is a key component of NAIRR, providing essential access to computing resources, datasets, and tools while laying the groundwork for a scalable and inclusive national AI infrastructure. NAIRR is also committed to supporting workforce development, ensuring that the next generation of AI practitioners, including RSEs, have the necessary training and tools to sustain U.S. leadership in artificial intelligence. By fostering collaboration between public and private sectors, NAIRR aims to strengthen national AI capabilities, drive economic growth, and enhance U.S. competitiveness in AI-driven industries.

4.2 Current Science Gateways in AI

From our science gateways community, we have provided high-impact examples of projects that have adopted AI solutions in their science gateway while acknowledging that this is not a complete list.

NAIRR Pilot Portal - *Advancing U.S. AI research*

- The NAIRR Pilot Portal [2] is an initial step toward democratizing access to AI computing, datasets, and tools for researchers, educators, and industry professionals. It is designed to test and refine AI resource-sharing strategies, offering a federated model for accessing computing power, open-source AI frameworks, and curated datasets. It is maintained and further developed by SGX3.
- AI Integration (Planned): Federal agencies are collaborating with government-supported and non-governmental partners to implement the Pilot as a preparatory step toward an eventual full NAIRR implementation. This pilot phase is critical in shaping the long-term vision of the NAIRR, ensuring that future iterations address key challenges in AI accessibility, interoperability, and resource allocation. Community engagement and feedback will be instrumental in defining its next steps.

nanoHUB - *Nanotechnology, materials science, and AI-powered simulations*

- nanoHUB [3] is a widely used science gateway that integrates AI-driven models for materials discovery, semiconductor simulation, and quantum computing applications. It provides an interactive environment where researchers can run simulations, use machine learning models for materials informatics, and visualize computational results without requiring direct HPC expertise.
- AI Integration: AI/ML models are used for materials prediction, process optimization, and device simulation to train the next generation of nanotechnology engineers and certify the training of industry workers already in the workforce. Provides a training recommender for the industry workforce to elevate recommendations for the next training to enhance their skills and knowledge.

DesignSafe - *AI-powered natural hazard research*

- DesignSafe [4] is an engineering science gateway integrating AI and machine learning models for disaster resilience and structural engineering research. It provides a science gateway interface for running AI simulations, processing large datasets related to hazards (e.g., hurricanes, earthquakes, floods, wildfires, etc.), and developing predictive models for infrastructure resilience.
- AI Integration: Uses deep learning for hazard prediction, damage assessment, and infrastructure monitoring.

Brainlife - *Neuroscience and AI-powered brain imaging*

- Brain-Life [5] is a cloud-based neuroscience science gateway that provides AI-based tools for brain imaging analysis, neural network modeling, and cognitive research. It enables researchers to process large-scale neuroimaging datasets using machine learning pipelines and computing resources.
- AI Integration: Incorporates AI models for MRI segmentation, brain connectivity analysis, and disease diagnosis.

Jetstream2 - *AI-powered cloud computing for research across disciplines*

- Jetstream2 [6] is a cloud-based science gateway and research computing platform that supports AI workloads across various domains, including bioinformatics, geosciences, and social sciences. It offers virtualized AI environments where researchers can deploy machine learning models, run Jupyter notebooks, and access scalable computing resources.
- AI Integration: Provides access to GPUs for deep learning and pre-configured AI software environments.

MLCommons Science Gateway - *AI benchmarking and machine learning for science*

- MLCommons [7] provides an AI science gateway to benchmark machine learning models for scientific applications. It evaluates and compares AI performance across different hardware architectures and optimizes ML workflows for research computing.
- AI Integration: Focuses on standardizing AI model performance metrics and creating reproducible AI experiments.

4.3 Challenges in AI Adoption

AI adoption is hindered by critical challenges that can be addressed through improved connections that science gateways, RSEs, and AI agents can support. One challenge restricting innovation in AI adoption is the lack of access and the barriers between data, models, and computing. Many universities, research institutions, and businesses struggle to access the computational power and datasets required to train and deploy AI models at scale. The concentration of AI resources among a few large corporations raises concerns about AI's adoption by all.

Additionally, the shortage of AI-skilled professionals, particularly RSEs, remains a barrier to novel development and deployment. Demand for AI talent far exceeds supply, making it difficult for government agencies, academic institutions, and small enterprises

to compete with large technology firms for skilled workers.

Another challenge lies in the fragmentation of AI ecosystems, where researchers, educators, and developers often operate in isolated environments with limited opportunities for collaboration. Many AI models, datasets, and tools are proprietary or siloed, creating barriers to interoperability and slowing the progress of AI innovation. Addressing these challenges requires a national effort to create more open, collaborative AI research environments.

4.4 Science gateways and workforce development as a solution

An extended AI infrastructure - currently NAIRR - has been proposed to address these challenges as a critical initiative to democratize AI access and drive innovation. Extended AI infrastructure aims to provide researchers, educators, and industry professionals with smooth access to AI models, datasets, and HPC resources, reducing barriers to entry for AI-driven discoveries and applications. By offering a federated approach to AI resource distribution, the AI infrastructure can foster an AI research ecosystem that ensures that advancements in AI benefit a broad range of stakeholders.

A key component of a national AI infrastructure's success will be the integration of science gateways and AI-powered research portals that simplify access to AI tools. These portals will lower technical barriers, making AI adoption more accessible for all users. Additional recommendations include training RSEs to grow the workforce around AI solutions and the development of these solutions. Thus, the growth of AI agents in science gateways and portals will be influenced by all industries' workflow enhancements.

At the heart of the mission, we need a centralized structure backed by the federal government to continue growing these efforts. Thus, the U.S. can create a more efficient AI research environment by enabling continued relations across all sectors to expand access to AI infrastructure, science gateways, and AI agents. These initiatives will bridge the gap between large technology firms and smaller institutions, fostering a more collaborative AI research community that accelerates economic growth, scientific progress, and national security advancements. Investing in AI accessibility, workforce development, and computational infrastructure will ensure that the U.S. remains at the forefront of AI innovation.

5 Challenges and Barriers to AI Leadership

Despite the rapid advancement of artificial intelligence, the U.S. faces significant challenges in maintaining its leadership in AI research, development, and deployment. Key barriers include infrastructure limitations, shared knowledge within the workforce, and AI-driven decision support systems. To be AI forward, we should address these challenges and enable all sectors to advance through AI adoption programming.

5.1 Infrastructure bottlenecks and fragmented AI resources

The U.S. is currently bottlenecked by the limited access and high cost of computing resources necessary for training and deploying AI models. AI applications, particularly

deep learning and foundation models, require computational time and resources to shape themselves into useful solutions. Many research institutions, innovative startups, and enterprises have limited access to the resources needed to train their innovative solutions. Thus, many solutions cannot receive enough support to be ready for adoption beyond initial testing.

In addition to computing limitations, datasets are fragmented and scattered across resources. Knowledgeable parties must hunt down the data to curate in one location to train the AI solution. Organizations maintain proprietary datasets that are difficult to share without secure connections or enclaves, stalling innovation in these sectors. Without a coordinated effort to establish standardized data-sharing protocols and federated AI infrastructures, AI research will remain fragmented, limiting the nation's ability to leverage AI for large-scale problem-solving.

5.2 Workforce and AI Knowledge Gaps

The U.S. needs to train and extend knowledge about AI to professionals in all sectors. Today, certificate programs are scattered across professional spaces in academia and industry—such as starter certificates on LinkedIn—but these trainings are scattered, and a professional has to navigate multiple systems to find all the training available or hear about the offerings through word of mouth. Designed AI training programs and digested programs in one portal to streamline workforce onboarding.

A national AI infrastructure can play a crucial role in filling these workforce gaps by providing AI training programs through science gateways and RSE-led initiatives. These programs can help train the next generation of AI researchers, developers, engineers, and adopters of AI in workflows.

5.3 The need for AI-driven decision support

As AI research and applications grow more complex, there is an increasing need for AI-driven decision-support systems to assist researchers, policymakers, and industry professionals. AI workflows involve multiple stages, from data preprocessing and model selection to training, evaluation, and deployment. Manually managing these processes can be time-consuming, error-prone, and inefficient.

AI agents can help optimize AI workflows by automating routine tasks, suggesting appropriate models, and generating insights from complex datasets. These intelligent systems can analyze large volumes of data, recommend best-fit AI models, and assist researchers in fine-tuning parameters, significantly reducing the time and effort required for AI experimentation.

Beyond research, AI agents can support policymakers by providing data-driven insights on AI governance, risk assessment, and ethical considerations. Automated decision-support tools can help government agencies evaluate the impact of AI policies, track AI adoption trends, and ensure compliance with AI safety standards.

AI agents should be integrated into the research infrastructure to fully realize the benefits of AI-driven decision support. This would allow users to leverage AI automation to optimize workflows, manage computing resources, and improve the efficiency of AI

development. Embedding AI agents within science gateways can further enhance accessibility, usability, and innovation in AI research.

6 Proposed Policy Actions

A strategic approach is necessary to maintain and enhance U.S. leadership in AI. The following policy actions will establish an AI ecosystem.

6.1 Establish a national AI infrastructure

Centralized AI infrastructure is critical to accelerating research and technological advancements. A key priority should be the federation of computing resources and datasets via science gateways, which would connect AI tools, computing power, and large-scale datasets that are available nationwide.

Investment in RSEs is essential to building and maintaining sustainable AI research software. Federal funding should support RSE-driven software development, guaranteeing that AI models and datasets are integrated seamlessly into research environments.

Integrating AI agents within AI research platforms will accelerate innovation by automating AI model selection, optimizing computing workflows, and assisting researchers in decision-making processes. AI agents can recommend suitable AI models, handle data preprocessing, and facilitate large-scale computational tasks, improving the overall efficiency of AI research.

6.2 Strengthen AI workforce development through AI infrastructure

A skilled workforce is the foundation of sustainable AI development, and targeted investments in AI education and career pathways are needed to meet growing industry demands. Funding for RSE education programs and career initiatives should be expanded to support the next generation of AI software developers, ensuring a steady supply of talent to maintain AI research infrastructure.

AI training programs should be integrated into science gateways to provide hands-on experience in AI development, computing resource allocation, and AI model optimization. These programs should be accessible to all in the existing workforce the next generation of the workforce. Thus, ensuring an AI capable workforce.

Additionally, AI agents should be leveraged as digital tutors and recommenders - as noted in the case of nanoHUB - to facilitate AI education and workforce training. AI-driven learning systems can personalize instruction, provide adaptive learning experiences, and automate assessment. AI agents not only help the learners but support the educators growing the curriculum as it rapidly advances to make AI education more scalable and effective. These AI agents will ensure that students and professionals can build expertise in AI.

6.3 Enhance AI infrastructure through science gateways

Science gateways are essential for expanding access to AI tools, datasets, and computational resources. AI-enabled research portals should be developed to serve as entry points for AI researchers, allowing them to seamlessly interact with AI computing frameworks, datasets, and pre-trained models.

AI agents should be integrated into these science gateways to streamline user workflows and development. For example, automating routine tasks around data preprocessing, model selection, and AI benchmarking. Automating these processes will allow focus on advancing AI applications rather than managing technical complexities.

6.4 Drive AI leadership through strategic federal investments

Sustained investment in AI infrastructure and workforce development requires federal funding allocations and strategic partnerships with the private sector. Federal agencies should allocate dedicated funding for integrating science gateways into AI research infrastructure to provide scalable access to AI tools, ensuring that computing resources and datasets are efficiently distributed.

Public-private partnerships will be crucial in advancing AI agent research and development for research, healthcare, and national security applications. These partnerships should support AI innovation while ensuring AI-driven solutions align with ethical and regulatory considerations.

A national RSE workforce initiative should also be established to provide career incentives, specialized training programs, and funding for AI software engineering roles. By investing in AI-focused RSE programs, the U.S. can develop a workforce capable of maintaining AI infrastructure and scaling AI applications for both public and private sectors.

6.5 Strengthen AI for national security and economic competitiveness

AI is critical in enhancing national security, intelligence, and economic resilience. Policy actions should focus on developing AI-powered intelligence and defense applications. These policies should be refined through the sectors as they integrate AI solutions, but through the use of science gateways and AI agents, policies can be adapted to think through the online environments to be deployed. By streamlining a common type solution for all sectors, enabling policies and access to also be streamlined. The use of science gateway secure enclaves can also be deployed for sensitive and restricted sectors.

AI agents should be deployed for cybersecurity applications, including real-time threat detection, automated response mechanisms, and AI-driven vulnerability assessments. These AI systems will help safeguard government and critical infrastructure networks from cyberattacks and AI-generated threats.

AI-enabled supply chain monitoring platforms should be developed to ensure the security and resilience of U.S. supply chains. AI models can predict supply chain disruptions, optimize logistics, and detect vulnerabilities in global trade networks,

reducing economic risks associated with international dependencies.

Expanding RSE roles in AI security and resilience projects will be essential for maintaining secure AI systems in defense, cybersecurity, and government operations. By ensuring that AI models are designed with security and transparency, the U.S. can prevent adversarial AI attacks and enhance the robustness of AI-driven decision-making.

6.6 Expand U.S. global AI leadership through AI infrastructure

To remain competitive on the global stage, the U.S. must expand its AI infrastructure as an international AI research hub. To do this, science gateways can be leveraged to foster collaboration, including with allied nations. Science gateways can be leveraged to connect AI partnerships globally, allowing researchers across borders to access shared datasets, AI tools, and computing resources.

Standardizing AI agent interoperability will be key to ensuring global AI system compatibility. Establishing international AI standards will promote seamless AI integration across different countries, facilitating joint research initiatives and cross-border AI applications.

International collaborations among RSEs should be encouraged to strengthen AI research and software sustainability. Establishing AI software development partnerships with research institutions worldwide will enhance AI model development, promote open-source AI advancements, and ensure that AI applications adhere to global best practices.

7 Implementation Roadmap

A phased implementation strategy is necessary to successfully deploy AI infrastructure, workforce development, and AI-driven decision support. This roadmap outlines short-term, medium-term, and long-term actions to launch the U.S. as an AI global economic leader.

7.1 Short-Term Actions (1-2 years)

The initial phase will focus on establishing pilot programs, deploying AI agents, and creating foundational workforce initiatives to accelerate AI adoption and research collaboration.

- Launch pilot science gateways that are individually sector focused for AI-powered research and model sharing to provide researchers, educators, and industry professionals with streamlined access to AI computing resources, datasets, and collaborative tools. These pilot gateways will be testbeds for optimizing AI infrastructure usability and accessibility.
- Deploy AI agents to support AI adoption within research institutions, industry collaborations, and federal agencies. AI agents will assist in model selection, data processing, and workflow automation, reducing technical barriers to AI integration.

- Create national RSE fellowships to attract and train research software engineers for AI research and development. These fellowships will support early-career RSEs and provide opportunities for hands-on experience in AI infrastructure maintenance, model optimization, and software engineering for AI applications.
- Develop AI training resources within science gateways to offer online courses, interactive tutorials, and hands-on AI development projects, ensuring researchers and professionals can efficiently leverage AI tools and computing resources.
- Establish initial public-private partnerships to align AI research with industry needs, focusing on collaborative projects integrating AI into sector forward secure enclaves (e.g. national security, healthcare, and industrial automation).

7.2 Medium-Term Actions (3-5 years)

Building on the success of short-term initiatives, the medium-term phase will focus on scaling AI computing infrastructure, automating AI workflows, and fully integrating RSEs into federally funded AI programs.

- Expand federated computing and data access across U.S. research institutions to enhance AI collaboration and standardize AI model-sharing frameworks. This expansion will increase accessibility to computing power for under-resourced research institutions and start-up AI-solution enterprises.
- Scale AI agent capabilities to automate large-scale research projects by enhancing AI-driven decision support, optimizing resource allocation, and providing advanced analytical insights. AI agents will also support policymakers in assessing AI policies, national security risks, and regulatory compliance.
- Integrate RSEs will become critical to AI workforce planning, contributing to AI software engineering, model reproducibility, and computing infrastructure optimization.
- Establish regulatory and security frameworks for AI adoption across critical sectors, focusing on AI governance, cybersecurity, and ethical AI development. These frameworks will help ensure safe AI deployment in data-sensitive and highly regulated sectors (e.g. defense, healthcare, and financial systems).
- Increase investment in AI education and workforce development through expanded AI fellowships, industry training programs, and university-led AI research initiatives, ensuring a steady pipeline of skilled professionals.

7.3 Long-Term Vision (5+ years)

The long-term goal is positioning AI infrastructure and science gateways as a global model for AI research collaboration, workforce development, and economic growth.

- Establish AI infrastructure and science gateways as an international research model by fostering AI research collaborations with allied nations, enabling cross-border AI research, and standardizing AI development frameworks globally.
- Integrating AI agents into national security, industrial automation, education, and research-driven industries will ensure that they enhance productivity across

sectors. AI-driven decision-making systems will optimize efficiency, resource management, and workforce augmentation.

- Institutionalize RSE roles as a cornerstone of the U.S. AI workforce by embedding RSEs within national AI research initiatives, federally funded projects, and private-sector AI research programs. This will ensure long-term sustainability and innovation in AI software infrastructure.
- Develop next-generation AI computing capabilities through investments in quantum AI computing, energy-efficient AI hardware, and secure AI cloud infrastructures, ensuring that the U.S. remains at the forefront of AI technological advancements.
- Promote U.S. leadership in AI governance and ethical AI development by setting global standards for AI safety, interoperability, and cybersecurity, ensuring responsible AI deployment worldwide.

This phased implementation leads to a strengthened AI infrastructure in the U.S., scales workforce development, and integrates AI-driven decision support across key sectors. Through pilot programs, infrastructure expansion, automation, and international collaboration, the U.S. can solidify its position as a global AI leader while ensuring AI technologies remain accessible, secure, and economically beneficial.

8 Conclusion

Artificial intelligence is a transformative force driving scientific discovery, economic growth, and national security. A strategic investment in AI infrastructure, workforce development, and decision-support systems is essential to maintain U.S. leadership in AI. This response has outlined key recommendations to enhance AI accessibility, optimize research efficiency, and strengthen the AI workforce, ensuring that AI innovation remains at the forefront of U.S. technological advancement.

Central to this vision is the expansion of AI infrastructure as the backbone of U.S. AI research and development, with science gateways providing seamless access to AI computing resources, models, and datasets. These gateways will enable AI innovation, ensuring that researchers, educators, and industry professionals have the tools they need to drive technological solutions.

AI agents will serve as workforce multipliers. Streamlining AI-driven discovery by automating routine tasks, optimizing computational workflows, and supporting decision-making across research, national security, and industry applications. By integrating AI agents into AI research portals and science gateways, researchers can accelerate model development, improve reproducibility, and enhance AI-driven problem-solving.

Equally critical is the role of RSEs in maintaining and optimizing AI infrastructure. RSEs ensure AI software is accomplished in the U.S.. Investing in RSE-led AI software development programs will strengthen the nation's AI workforce, ensuring that expertise

in AI engineering, computing resource management, and security remains a national priority.

Federal support for scalable AI platforms is needed to secure the U.S.'s global AI leadership. This includes funding for federated AI computing systems, AI education and training initiatives, and the expansion of public-private partnerships to drive AI adoption in critical sectors. By fostering innovation through government, academic, and industry collaboration, the U.S. can set the global standard for AI research, development, and ethical deployment.

To future-proof U.S. AI infrastructure, immediate investment is needed in AI-powered science gateways, AI agent technology, and RSE workforce development. These investments will ensure that AI remains accessible, scalable, and aligned with national security and economic priorities. The U.S. can set the global standard for AI research, development, and ethical deployment by fostering innovation through government, academic, and industry collaboration.

Glossary of Key Terms

Artificial Intelligence (AI) – A branch of computer science focused on developing systems that can perform tasks typically requiring human intelligence, such as learning, reasoning, problem-solving, and natural language understanding.

AI Agent – An intelligent system that can automate decision-making, optimize workflows, and assist in AI-driven research by handling tasks like data analysis, model selection, and computational resource management.

AI Infrastructure – The hardware, software, and organizational frameworks required to support AI research and development, including computing resources, AI models, and data-sharing platforms.

AI Model – A trained machine learning or deep learning algorithm for tasks such as language processing, image recognition, and data predictions.

Deep Learning – A subset of machine learning that uses artificial neural networks to process large amounts of data and perform complex tasks such as image recognition and speech processing.

Federated Learning – A machine learning approach that allows AI models to be trained across multiple decentralized devices or servers without sharing raw data, enhancing privacy and security.

Foundation Model – A large-scale AI model trained on vast datasets that can be adapted for various tasks, such as natural language understanding, image generation, or scientific research.

High-Performance Computing (HPC) – The use of powerful computing systems to process and analyze large datasets at high speeds crucial for AI research and simulation-based studies.

Machine Learning (ML) – A subset of AI that enables computers to learn patterns and make predictions from data without being explicitly programmed.

NAIRR (National Artificial Intelligence Research Resource) – A U.S. federal initiative designed to democratize access to AI resources, providing researchers and educators

with computing power, datasets, and AI tools.

Natural Language Processing (NLP) – A field of AI that enables machines to understand, interpret, and generate human language, used in applications such as chatbots, translation, and sentiment analysis.

Neural Network – A computational model inspired by the human brain, consisting of layers of interconnected nodes that process and learn from data.

Open-Source AI Tools – AI software and frameworks that are freely available for public use, modification, and collaboration, fostering innovation and accessibility in AI research.

Research Software Engineer (RSE) – A professional who specializes in developing, maintaining, and optimizing software for scientific research, including AI-driven applications.

Science Gateway – A web-based platform that provides researchers and educators with user-friendly access to computational tools, AI models, and shared datasets for scientific discovery.

Secure Cloud Environment – A computing environment that provides AI researchers access to cloud-based resources while ensuring data privacy, security, and compliance with regulations.

References

- [1] Gesing, S., Stirm, C., Zentner, M., Dahan, M., & Hayden, L. 2023. SGX3: Novel Concepts to Enhance Knowledge and Extend the Community Around Science Gateways. Science Gateways 2023 (SG23), Pittsburgh, PA. <https://doi.org/10.5281/zenodo.10034892>
- [2] <https://nairrpilot.org/>
- [3] Gerhard Klimeck, M. McLennan, Mark Lundstrom, George Adams III, (2008), "nanoHUB.org - online simulation and more materials for semiconductors and nanoelectronics in education and research", 8th IEEE Conference On Nanotechnology, 2008. Nano, Nanotechnology, 2008. NANO'08. 8th IEEE Conference on, : pg: 401-404, IEEE, 08
- [4] Rathje, E., Dawson, C. Padgett, J.E., Pinelli, J.-P., Stanzione, D., Adair, A., Arduino, P., Brandenburg, S.J., Cockerill, T., Dey, C., Esteva, M., Haan, Jr., F.L., Hanlon, M., Kareem, A., Lowes, L., Mock, S., and Mosqueda, G. 2017. "DesignSafe: A New Cyberinfrastructure for Natural Hazards Engineering," ASCE Natural Hazards Review, doi:10.1061/(ASCE)NH.1527-6996.0000246
- [5] Hayashi, S., Caron, B.A., Heinsfeld, A.S. et al. brainlife.io: a decentralized and open-source cloud platform to support neuroscience research. Nat Methods (2024). <https://doi.org/10.1038/s41592-024-02237-2>
- [6] Zonca, A., Weakley, L. M., & Standish, M. (2023, October 30). Interactive distributed computing with Jupyter, Python and Dask on Jetstream2 for Science Gateways. Science Gateways 2023 (SG23), Pittsburgh, PA. Zenodo. <https://doi.org/10.5281/zenodo.10041240>.
- [7] <https://mlcommons.org/>